



Creando Oportunidades

Internal Model Method for Counterparty Credit Risk

Methodology

Madrid, April 2026

Index

Index.....	2
Versions.....	4
Glosario de siglas y acrónimos.....	5
1. Introduction.....	7
2. Calculation Scope.....	7
3. Carving-out.....	7
4. Benchmarking.....	8
4.1. MtM Adjustments.....	8
5. General Model.....	9
5.1. Exposure At Default.....	9
5.1.1. Modelled part.....	10
5.1.1.1. Grid.....	10
5.1.1.2. Effective EPE.....	11
5.1.1.3. CVA.....	11
5.1.1.4. Alpha.....	12
5.1.1.5. Normal and stressed calculations.....	12
5.1.2. Non-modelled part.....	12
5.2. Risk Weights.....	12
5.2.1. Correlations for IRB portfolios.....	13
5.2.2. Maturity Adjustment.....	13
5.2.2.1. Effective Maturity.....	14
6. Risk Factors.....	16
7. Valuation Issues.....	17
8. Collateral and Margin Period of Risk.....	17
8.1. Collateral Allocation.....	17
8.1.1. Allocation of Initial Margin and CSA parameters.....	18
8.1.2. Allocation of Variation Margin (VM).....	18
8.2. MPoR.....	19
8.3. Dynamic Initial Margin.....	20
8.4. Variation Margin Model: Settlement-Adjusted Path Dependant.....	23
8.4.1. Call Dates.....	24
8.4.2. MtM at non-simulated call dates (Brownian-bridge).....	25
8.4.2.1. Variance Covariance (VCV) Matrix.....	26
8.4.3. Initial Variation Margin Balance (IVMB).....	28
8.4.4. Variation Margin Required (VMR) at a call date.....	30
8.4.5. Margin call (MC) and recursive update (algorithm).....	31
8.4.6. Variation Margin Smooth and Variation Margin Full.....	33
8.4.7. Collateralised exposure.....	33
8.4.8. Settled-to-Market convention.....	34

8.5. Spikes and Smooth Exposure.....	35
8.5.1. CLS.....	36
8.5.2. Swap Agent.....	36
8.6. Default Management Process.....	36
8.7. Penalties in the MPoR.....	37
8.8. Haircuts.....	37
9. Backtesting.....	38
9.1. Risk Factors Backtesting.....	38
9.2. Portfolio Backtesting.....	39
9.3. Global Backtesting, traffic lights.....	40
10. Stress Testing.....	40
10.1. Extra Stressed Period Search.....	41
10.2. Lehman and Covid.....	41
10.3. Forward Looking Implicit.....	42
10.4. Forward Looking ICAAP.....	42
10.5. Reverse Stress Testing.....	43
11. Wrong Way Risk.....	43
11.1. General WWR.....	43
11.1.1. Methodological Framework.....	44
11.1.1.1. Credit State Evolution in RiskWatch.....	44
11.1.1.2. Systemic Credit Scalars.....	44
11.1.1.3. Simulation and Calibration.....	44
11.1.2. Implementation and Practical Considerations.....	45
11.1.2.1. Specific Weights and IRC Model Integration.....	45
11.1.2.2. Granularity Decision.....	45
11.1.2.3. Exposure Assumption: Smooth vs Full.....	45
11.1.3. Monitoring and Governance.....	45
11.2. Specific WWR.....	46
12. RNIEPEs.....	46
13. CORE.....	47
14. Expert criteria outside the model.....	47

Versions

Version	Date	Owner	Description	Approval
1.0	25/11/2025	GMRU COE	Initial version	GRMC
2.0	30/01/2026	GMRU COE	RNIEPE validation Added new MonteCarlo error analysis	VI GRM Data & Analytics (Johan Gunnesson)
2.1	31/03/2026	GMRU COE	Incorporated methodological details from Vendor documentation into main document	GMRU - Admission & CCR Measurement (Soledad Seivane) GRM Data & Analytics (Johan Gunnesson)
2.1.1	08/04/2026	GMRU COE	Update exposure formulae and DIM scaling factor to correct erratas	GRM Data & Analytics (Johan Gunnesson) GMRU - Admission & CCR Measurement (Soledad Seivane)

Glosario de siglas y acrónimos

*Adjustment*_{mat}: maturity adjustment

CCR: Counterparty Credit Risk

CF: Cash Flow

CLS: Continuous Linked Settlement

CRM: Credit Risk Mitigation

CRR: Capital Requirements Regulation

CSA: Credit Support Annex

CVA: Credit Valuation Adjustment

DIM: Dynamic Initial Margin

DMP: Default Management Process

DVA: Debt Value Adjustment

EAD: Exposure at Default

EE: Expected Exposure

EPE: Expected Positive Exposure

FRTB: Fundamental Review of the Trading Book

gWWR: General Wrong Way Risk

HN: Historical Normal calibration period

HS: Historical Stressed calibration period

IA: Independent Amount

ICAAP: Internal Capital Adequacy Assessment Process

IM: Initial Margin

IMM: Internal Model Method

IRB: Internal Ratings-Based approach

ISDA: International Swaps and Derivatives Association

KCCR: Capital requirements for Counterparty Credit Risk

LGD: Loss Given Default

M: effective maturity

MPoR: Margin Period of Risk

MTA: Minimum Transfer Amount

MtM: Mark To Market

NPV: Net Present Value

PCA: Principal Component Analysis

PD: Probability of Default

PFE: Potential Future Exposure

RNIEPEs: Risks-Not-In-Expected-Positive-Exposure

RW: Risk Weights

SA-CCR: Standardised Approach for Counterparty Credit Risk

SFT: Securities Financing Transaction

SIMM: Standard Initial Margin Model

SME: Small and Medium Enterprise

SNS: Synthetic Netting Sets

sWWR: Specific Wrong Way Risk

TH: Threshold

VaR: Value at Risk

VM: Variation Margin

WWR: Wrong Way Risk

1. Introduction

The aim of this document is to describe how exposures for regulatory counterparty credit risk capital charge in BBVA S.A. are calculated and applied to derivatives in BBVA's portfolios.

The methodology presented in this document applies to the following models defined within the Internal Model Method (IMM) framework:

- IMM Counterparty - Exposures engine
- IMM Counterparty - gWWR
- IMM Counterparty - Backtesting
- IMM Counterparty - Stress Testing
- IMM Counterparty - RNIEPE

2. Calculation Scope

The scope of this calculation is BBVA S.A.'s derivatives portfolio, specifically those listed in CRR Annex II.

Securities lending and repos are not subject to the IMM; their calculation will be based on the Credit Risk Mitigation (CRM) model.

3. Carving-out

As permitted by Capital Requirements Regulation, institutions may calculate exposures from certain trades using the Internal Model Method (IMM), while applying the Standardised Approach for Counterparty Credit Risk (SA-CCR) to others (Article 283 CRR). In such cases, netting sets containing both types of trades must be split into two Synthetic Netting Sets (SNS).

Trades eligible for IMM treatment are those for which full revaluation is applied and no significant deficiencies exist in the IMM pricing model that would lead to material deviations from the actual mark-to-market (MtM). This typically includes interest rate, foreign exchange, and certain equity derivatives. All remaining trades, which lack reliable valuation inputs or for which the IMM simulation fails to operate properly, are treated under SA-CCR.

In the case of structured trades, if any component cannot be appropriately valued under IMM due to deficiencies or missing inputs, the entire structure must be assigned to SA-CCR. This strict rule ensures prudence and consistency in exposure measurement.

The separation of netting sets also necessitates an appropriate **collateral allocation** process to distribute collateral fairly between the synthetic netting sets (see Section [8.1, Collateral Allocation](#)).

4. Benchmarking

Institutions are expected to perform a benchmarking exercise comparing the IMM-based valuation of trades with the Front Office valuation.

All trades showing material differences—measured using absolute and relative deviations, as well as differences between mark-to-market (MtM) and nominal values at the initial time point (t_0)—must be considered for exclusion from IMM and instead calculated under SA-CCR, applying the carving-out described in the previous section.

However, in cases where such trades are not excluded from IMM, a mandatory MtM Adjustment must be applied to the client exposure in order to compensate for the valuation alert detected during benchmarking. This ensures that the exposure used in capital calculations remains prudent even if the trade is retained within the IMM scope.

In the case of structured products, if any component fails the benchmarking criteria, the entire structure must be assigned to SA-CCR. This conservative treatment ensures the integrity and reliability of internal model estimates across all components of complex products.

For more details, see this document:

[Benchmarking Manual](#)

4.1. MtM Adjustments

As part of the benchmarking process, the mark-to-market (MtM) values produced by the IMM pricing libraries are compared with the official Front Office benchmark valuations validated through the Independent Price Verification (IPV) framework.

In this context, **material differences** refer to valuation deviations that exceed the predefined materiality thresholds established in the IMM benchmarking framework. These thresholds are assessed using three complementary indicators: the absolute difference between the IMM MtM and the benchmark MtM, the relative difference expressed as a percentage of the benchmark MtM, and the difference expressed as a percentage of the trade notional. A trade is flagged as an alert only when the thresholds associated with all three indicators are breached simultaneously.

When material differences are detected, trades are assessed to determine whether they should be excluded from the IMM scope and treated under SA-CCR, as described in Section 3 (Carving-out).

In cases where the trade remains within the IMM scope despite the valuation difference, a prudential MtM adjustment is introduced in order to ensure that exposures are not understated due to model valuation discrepancies.

The **adjustment** is defined as the positive difference between the benchmark valuation and the IMM valuation at the reference date:

$$\Delta MtM_i(t_0) = MtM_i^{benchmark}(t_0) - MtM_i^{IMM}(t_0)$$

Only positive differences are considered, so that the adjustment is one-sided and conservative.

Individual trade differences are aggregated at the most granular legally meaningful level available (CSA, Master Agreement or counterparty level). The aggregated difference is then incorporated into the exposure profile as an additive overlay that decays as the affected trades approach maturity.

This mechanism ensures that the exposure used for regulatory capital purposes remains aligned with benchmark valuations while preserving the IMM treatment of the affected trades.

5. General Model

The model used to compute regulatory capital for counterparty credit risk aims to capture, over a one-year time horizon:

- losses due to counterparty defaults in derivatives
- losses due to declines in credit quality

The regulatory capital requirement for CCR is calculated using the following formula:

$$KCCR = EAD * RW$$

Where:

- EAD is the Exposure at Default, as determined under the IMM
- RW is the applicable risk weight, based on the counterparty's credit quality and other relevant factors (as per CRR Article 153 for IRB exposures, Articles 114-122 for exposures under the Standardized approach).

The following sections describe the methodology used to compute each component in detail.

5.1. Exposure At Default

The key difference between the modelled and non-modelled parts lies in the calculation of Exposure at Default (EAD), which is detailed in the following sections.

5.1.1. Modelled part

For exposures treated under the Internal Model Method (IMM), the Exposure at Default (EAD) is calculated using the effective Expected Positive Exposure (effective EPE), in accordance with Article 284(5) CRR. The formula is:

$$EAD = \max(\alpha \cdot \text{effective EPE} - CVA, 0)$$

Where:

- effective EPE is determined as per Article 284(4) and (5)
- α (alpha) is a regulatory multiplier, typically set to 1.4, unless a lower value is approved under Article 284(4)
- CVA is the credit valuation adjustment component deducted from exposure to avoid double counting, as referenced in Article 273(6).

All calculations involving modelled trades are performed **at the counterparty level** (Article 272(4) CRR), although original netting agreements are respected and we are not allowing any offsetting of positive and negative exposures coming from different netting sets.

5.1.1.1. Grid

The Expected Positive Exposure (EPE) must be computed over a set of future horizons spanning one year from now, in accordance with Article 284(6) of the CRR. Using daily timesteps up to a year requires significant computational effort, which does not substantially improve the reliability of the effective EPE compared to using a more limited set of timesteps.

This topic has been analyzed in detail in the following studies:

[IMM Exposure Numerical Error Study](#) → initial analysis performed in 2020.

[IMM Exposure Time Grid Numerical Analysis](#) → updated study conducted in 2025, which includes an assessment of the impact of time grid discretization and the Brownian Bridge interpolation.

To balance accuracy and computational efficiency, a non-uniform time grid is applied: time steps are daily in the near term (to capture short-term exposure dynamics accurately), and become gradually more spaced (e.g., weekly, then monthly, and finally annual intervals) as the time horizon extends beyond the first few weeks. Currently, the model employs 34 temporal nodes within the first year, and a total of 100 simulated time steps overall (101 if the t_0 reference date is included), including those beyond the one-year horizon, which are specifically used to calculate the effective maturity. Also, the 1y node impacts effective EPE (see next section) and is placed a few days later than the one-year mark, so the effective EPE is based on 35 temporal nodes, including t_0 .

Finally, it should be noted that the time step grid is adjusted to weekends and holidays, but the simulation model works in calendar days, implying generated volatility over weekends for instance.

5.1.1.2. Effective EPE

The effective EPE is computed as follows:

$$effective\ EPE = \frac{1}{min(1\ year,\ maturity)} \sum_{t_i=1}^{min(1\ year,\ maturity)} \Delta_{t_i} * effective\ EE(t_i)$$

where

$$effective\ EE(t_i) = \max_{t_j \leq t_i} EE(t_j)$$

at each time t_i ,

$$\Delta_{t_i} = t_i - t_{i-1}$$

and $EE(t_i)$ is calculated at a counterparty level as explained in [Collateralized exposure](#).

For the term “1Y” in the formula, it is considered the number of days between the reference date and one year later (365 days). For the exact one-year value, the last partial interval is weighted proportionally to the number of days between the last simulation date before one year and one year, and the exposure value from the next simulation date is used¹.

In the case where the exposure profile has only the first date with a nonzero value, the effective EPE should be equal to that value.

5.1.1.3. CVA

The regulation allows the contingent unilateral CVA to be deducted, since the DVA is subtracted from the capital base.

Since CVA is calculated at the counterparty level, an allocation between the modelled and unmodelled parts must be performed. This allocation is based on an EAD calculated as if the entire portfolio were under the SA-CCR model (**EAD attribution**). The EAD of each trade serves as a weight for that trade.

All details can be found in the following document:

[Cálculo del vencimiento efectivo en KCCR regulatorio, y reparto por operación para propósitos de reporting](#)

¹ In the timestep grid we use, the 1y node is placed a few days after the 365 day mark, so this exposure node will correspond to the 1y node.

5.1.1.4. Alpha

Institutions may apply for permission to use an internally calibrated alpha under certain conditions, but this is subject to strict validation and supervisory approval.

Alpha parameter is not modelled internally and therefore is assigned the regulatory prescribed value, which is at least 1.4, in accordance with Article 284(4)(a) of the CRR. This conservative floor accounts for potential underestimation of effective EPE due to model limitations or risk not captured by simulation.

5.1.1.5. Normal and stressed calculations

Two separate counterparty credit capital calculations are performed for modelled trades:

- Normal (Current) Calculation (HN): using the most recent calibration period (which in our model is taken to be the last three years),
- Stressed Calculation (HS): using a historical stressed calibration period.

The capital charge for the modelled component is determined as the maximum of the two results.

5.1.2. Non-modelled part

For trades treated under the standard approach, SA-CCR calculations apply. Accordingly, exposure aggregation is performed at the level of **Synthetic Netting Sets (SNS)**.

Regarding CVA, the portion of CVA attributable to the standardised section is further allocated among the individual netting sets within that section. All details of this calculation can be found in the following document:

[Cálculo del vencimiento efectivo en KCCR regulatorio, y reparto por operación para propósitos de reporting](#)

5.2. Risk Weights

For Internal Ratings-Based (IRB) portfolios, risk weights (RW) are calculated using the regulatory formula established in Article 153 of the CRR:

$$RW = LGD * [\Phi(\frac{1}{\sqrt{1-\rho(PD)}} (\Phi^{-1}(PD) + \sqrt{\rho(PD)} * \Phi^{-1}(0.999))) - PD] * Adjustment_{mat}$$

where:

- Φ denotes the cumulative distribution function of the standard normal distribution
- $\Phi^{-1}(\cdot)$ denotes the inverse CDF (quantile function) of the standard normal distribution

- PD is the Probability of Default, and LGD is the Loss Given Default, both approved under the IRB approach
- ρ is the asset correlation
- $Adjustment_{mat}$ is the maturity adjustment factor

For non-IRB portfolios, standardized approach risk weights are applied (CRR Part 3, title II, chapter 2).

5.2.1. Correlations for IRB portfolios

In calculating the correlation used in risk weight (RW) formulas, a distinction is made between different types of counterparties. The correlation depends on the counterparty's probability of default (PD) and, for Small and Medium Enterprises (SMEs), also on their sales level. The formulas are as follows:

— Large or unregulated banks

$$\rho(PD) = 1.25 \cdot \left(0.12 \frac{1-e^{-50PD}}{1-e^{-50}} + 0.24 \cdot \left(1 - \frac{1-e^{-50PD}}{1-e^{-50}} \right) \right) = 0.15 \frac{1-e^{-50PD}}{1-e^{-50}} + 0.3 \cdot \left(1 - \frac{1-e^{-50PD}}{1-e^{-50}} \right)$$

— Rest of the banks / Sovereigns / Large Corporates / Public Sector Entities / Others, Wholesale

$$\rho(PD) = 0.12 \frac{1-e^{-50PD}}{1-e^{-50}} + 0.24 \cdot \left(1 - \frac{1-e^{-50PD}}{1-e^{-50}} \right)$$

— SMEs

$$\rho(PD) = 0.12 \frac{1-e^{-50PD}}{1-e^{-50}} + 0.24 \cdot \left(1 - \frac{1-e^{-50PD}}{1-e^{-50}} \right) - 0.04 \cdot \left(1 - \frac{S-5}{45} \right), 5 \leq S \leq 50$$

where S is the total annual sales of the SME, expressed in millions of euros, with the adjustment applicable for $5 \leq S \leq 50$.

5.2.2. Maturity Adjustment

This adjustment included in risk weights is intended as a penalty for long-term trades, which are considered to carry a higher risk of credit quality deterioration (i.e. risk of rating migration), and thus lead to increased capital requirements.

The maturity adjustment is defined as:

$$Adjustment_{mat} = \frac{1+(M-2.5)*b}{1-1.5b}$$

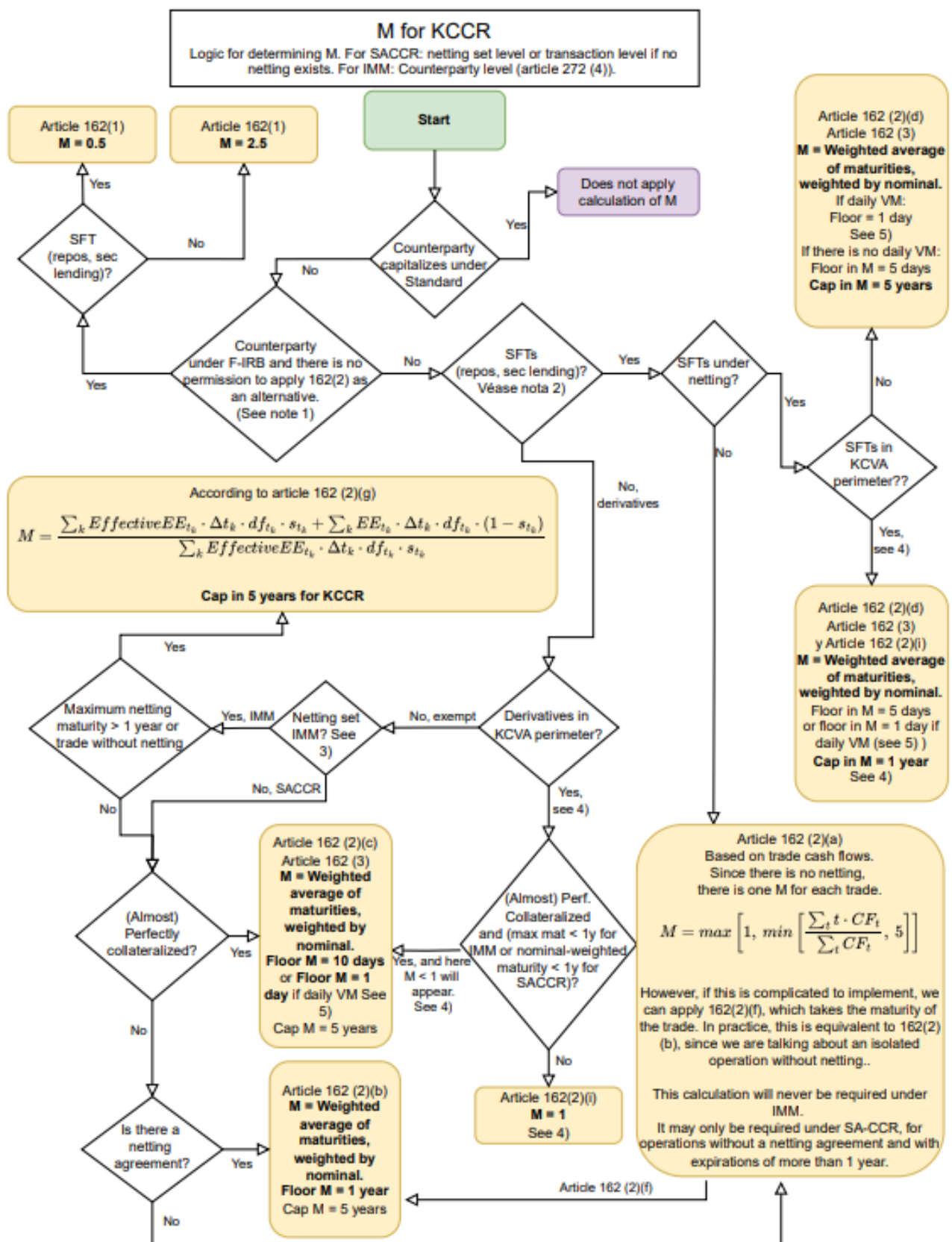
where

$$b = [0.11852 - 0.05478 * \ln(PD)]^2$$

and M is the effective maturity, the calculation of which is detailed in the following section.

5.2.2.1. Effective Maturity

The following diagram summarises the full calculation of the effective maturity parameter, which can be understood as a form of duration.



Notes: 1) CRR III makes it more likely that M calculations under paragraph 162(2) will be allowed, even if the exposure is under Foundation.
2) This diagram assumes that SFTs are outside the scope of the IMM. If SFTs are incorporated into the IMM, this logic changes.
3) When there is no netting, the isolated trade is considered a Netting Set (NS) under SACCR. Under IMM, the entity level is treated as a single NS.
4) Under CRR III, it is clear that with KCVA trades (basic or SA), there is a cap in KCCR for M in 1 year. Under CRR II, it requires A-CVA and supervisory authorization. Furthermore, under CRR II, it is unclear whether M = 1 is a cap or if M = 1 is fixed.
5) Article 162(3) allows the M floor for collateralized trades to be lowered to 1 day if it is demonstrated that there is daily margin and that the collateral can be liquidated or used to offset a default or failure to post collateral.

For the purposes of this calculation, Article 162(2)(a) of the CRR is not applied. Instead, weighted average maturity based on notional amounts is used, in accordance with Article 162(2)(f). This approach is equivalent to that described in Article 162(2)(b).

It should be noted that the correct application of break clauses contributes to reducing exposure, which may in turn lead to a decrease in the effective maturity parameter (M). Early termination of a contract depends on the nature of the **break clause**: if it is mandatory, the exposure is limited to the break date; if it is optional, exposure may be reduced provided there is a margin agreement in place and a positive legal opinion confirming the enforceability of the clause. Furthermore, break clauses falling within a short-term threshold prior to the valuation date are disregarded for exposure calculation purposes, and break clauses for SMEs are disregarded.

6. Risk Factors

BBVA's scenario generator manages four categories of risk factors: interest rates, inflation, foreign exchange (FX), and equity.

- For interest rates, the Hull-White two-factor model is used to capture the dynamics of the term structure.
- For inflation, both the Hull-White two-factor model and the Jarrow-Yildirim model are employed, to appropriately capture the joint dynamics of nominal rates, real rates, and inflation indexes.
- For equity and FX, the Extended Black-Scholes model is applied to represent the stochastic evolution of stock prices and volatilities.

During the calibration phase, the scenario generator is fitted to **historical market data**. In accordance with Article 292 CRR, for interest rates (nominal and real), calibration is performed using a **simulated annealing** optimisation algorithm. Two calibrations are conducted:

- One using a three-year rolling window, representing **current conditions**;
- a fixed three-year window representing a **stressed calibration**.

Each risk factor category is first calibrated independently. Subsequently, interdependencies between risk factors are captured through **calibration of the correlation structure**, in line with Article 284 CRR, which demands appropriate modelling of joint evolutions of market variables.

Due to the high dimensionality of the problem, **Principal Component Analysis (PCA)** is applied to reduce dimensionality while retaining 99% of the total explained variance.

Scenario paths are generated using the **Mersenne Twister** algorithm, a widely recognised pseudorandom number generator characterised by:

- an extremely long repetition period ($2^{19937} - 1$)
- the ability to pass numerous standard statistical randomness tests

- High performance in large-scale simulations.

Its capacity is more than sufficient to simulate 2,000 paths, each with approximately 2,000 risk factors and 100 timesteps.

Full technical details on the scenario generator are documented in:

[Generador de escenarios](#)

An in-depth analysis of Monte Carlo error related to the number of paths can be found in:

[IMM Exposure Numerical Error Study](#) → This study was originally conducted in 2020.

[IMM Exposure MonteCarlo Error Analysis](#) → A new analysis has been carried out following the methodology established in the last version of the ECB guidelines.

7. Valuation Issues

The pricing models applied in the IMM framework do not always coincide with those used for official valuations. This discrepancy often arises because counterparty credit risk calculations require certain approximations and model simplifications to ensure computational feasibility and efficiency.

Algorithmics' pricing libraries have been widely used for many years within current Market Risk models and have undergone extensive analysis and validation by the Internal Validation team. Nonetheless, benchmarking exercises are conducted regularly to ensure that there are no significant discrepancies between IMM valuations and Front Office valuations.

8. Collateral and Margin Period of Risk

8.1. Collateral Allocation

The separation into synthetic netting sets requires a collateral allocation. In this context, Initial Margin and CSA parameters (Thresholds, Independent Amounts, Minimum Transfer Amounts) are allocated according to the SA-CCR PFE add-on (SACCR add-on is first calculated at the operation level and then aggregated).

However, Variation Margin is allocated based on mark-to-market (MtM) values: the MtM amount is first assigned to each synthetic netting set, assuming that each set is either fully collateralised above the counterparty threshold (TH) and Minimum Transfer Amount (MTA), or below the user TH and MTA. Any resulting excess or shortfall is then allocated using the SA-CCR add-on.

8.1.1. Allocation of Initial Margin and CSA parameters

Initial Margin (IM) and CSA parameters (Thresholds, Independent Amounts, Minimum Transfer Amounts) are allocated across Synthetic Netting Sets (SNS) in proportion to each SNS contribution to the SA-CCR **PFE AddOn**. This guarantees that inputs intended to cover potential future exposure (PFE) are distributed according to the regulatory metric that represents PFE.

For a CSA that is split into two SNS (a simulated SNS s and a non-simulated SNS n), let $AddOn_s$ be the sum of all SA-CCR AddOns for all trades in the simulated SNS and $AddOn_n$ the sum of SA-CCR AddOns for all trades in the non-simulated SNS. The allocation factor for IM assigned to the simulated SNS is:

$$\rho_{IM} = \begin{cases} \frac{|AddOn_{IM,s}|}{|AddOn_{IM,s}| + |AddOn_{IM,n}|}, & \text{si } AddOn_{IM,s} \neq 0 \text{ or } AddOn_{IM,n} \neq 0 \\ \frac{1}{2}, & \text{otherwise} \end{cases}$$

The IM amounts assigned to the simulated and non-simulated SNS are then:

$$IM_s = \rho_{IM} \cdot IM_{total}, \quad IM_n = (1 - \rho_{IM}) \cdot IM_{total}$$

The same allocation factor is applied to the rest of the CSA parameters (Thresholds, Independent Amounts and Minimum Transfer Amounts).

8.1.2. Allocation of Variation Margin (VM)

First, the amount of VM required to fully collateralise the position is assigned to each SNS (above Threshold + MTA):

$$VM_s^{FullyColat} \text{ and } VM_n^{FullyColat}$$

Then, the residual VM is computed:

$$VM_{residual} = VM_{total} - VM_s^{FullyColat} - VM_n^{FullyColat}$$

Finally, a proportion of the residual VM is assigned to each SNS:

$$\rho_{VM} = \begin{cases} \frac{|AddOn_{VM,s}|}{|AddOn_{VM,s}| + |AddOn_{VM,n}|}, & \text{si } AddOn_{VM,s} \neq 0 \text{ or } AddOn_{VM,n} \neq 0 \\ \frac{1}{2}, & \text{otherwise} \end{cases}$$

ρ_{VM} is the proportion of the residual VM that is allocated to the simulated SNS; $1 - \rho_{VM}$ is the proportion of the residual VM allocated to de non-simulated SNS.

8.2. MPoR

The Margin Period of Risk (MPoR) in our implementation of the model is defined as:

$$MPoR = \max(Lag_VM, Lag_TS) + curePeriod$$

where:

- Lag_VM is the **margin lag**, i.e., the period during which it is assumed that only the non-defaulting party makes Variation Margin payments. As explained below, in our model margin lag is set to zero.
- Lag_TS is the **settlement lag**, i.e., the period during which only the non-defaulting party makes trade settlement flow payments
- **curePeriod**, i.e., number of days between the time that the user stops making any payments (margin and trade settlement) and the time the portfolio is hedged and/or liquidated.

It should be noted that these parameters are set in calendar days, since the exposure calculation in general is in calendar days. This means that the regulatory minimum MPoR in our system is set to 14 days.

We are aware that the above expression represents a departure from the regulatory MPoR definition in the case of a callPeriod greater than 1. A more complete approach would be to express the MPoR as:

$$MPoR = \max(Lag_VM, Lag_TS) + curePeriod + callPeriod - 1$$

which is consistent with the regulatory formulation

$$MPoR = F + N - 1$$

where **N** denotes the margining frequency (call period) and **F** represents the time to close-out following default. This is indeed the formula used for SA-CCR calculations.

The callPeriod is partially taken into account in the Variation Margin Model, as described in section 8.4. It should also be noted that the typical CSA agreement implies a callPeriod of 1 day (and is mandatory for financial counterparties under EMIR), in which case the applied formula is correct.

During the MPoR, the model assumes an asymmetric behaviour between the defaulting and the non-defaulting party. In particular, while the defaulted counterparty is assumed to cease payments, the institution may still be contractually required to honour trade settlement cash flows. As a consequence, trade-related cash flows paid by the institution to the counterparty during the MPoR result in upward jumps in the exposure time profile, referred to as **spikes** (see Section [8.5. Spikes and Smooth Exposure](#)).

The MPoR exposure profile is impacted by the existence and enforceability of early termination provisions, such as **break clauses**. When early termination is expected to occur –subject to a valid margin agreement and a positive legal opinion– the exposure horizon may be shortened accordingly, thereby reducing the formation of spikes that would otherwise result from assuming the transaction continues to full contractual maturity.

8.3. Dynamic Initial Margin

Initial margin (IM) is an important mitigant in CCR exposure and significantly reduces exposure for affected counterparties (small exposures are not required to be subject to Initial Margin requirements). Apart from variation margin an additional collateral buffer is provided to cover MtM variations during a specified period of time (10 days for instance) and at a specified confidence level (99% for instance).

Initial margin is normally posted by both BBVA and the counterparty, but is held by a 3rd party so that the interchange is not cancelled out. That way, the default of the counterparty will not affect the IM posted by BBVA, but the IM posted by the counterparty can help mitigate the exposure BBVA has to the counterparty.

Conceptually, Initial Margin represents MtM variation at a given level of confidence, as stated above, but in practice several different methodologies are used, depending on the type of counterparty. For OTC derivatives, the standard methodology applied is an industry standard defined by the International Swaps and Derivatives Association (ISDA) called SIMM (Standard Initial Margin Model). This model proxies MtM variability of a portfolio by a parametric VaR based on sensitivities, similar to how the FRTB Standardized Approach works for market risk.

A problem for CCR modelling is that it is not enough to be able to calculate IM at $t=0$, but instead we need a dynamic estimate which projects future values of IM, and takes into account how deals are maturing in the portfolio. Ideally we would project future values of ISDA SIMM, but this is computationally very difficult since it would require calculating a large set of MtM sensitivities in future time steps and simulations. So instead we will adopt a proxy, which we rescale to coincide with the $t=0$ Initial Margin value, as obtained from the official IM calculation systems (Calypso at the time of writing this document).

The proxy consists in calculating standard deviations of MtM variations between two timesteps and then rescale this standard deviation to the time horizon and the confidence level assumed by the

initial margin assuming $\sqrt{t}$ dependence with time and a normal distribution of MtMs. As mentioned earlier, this IM estimate is then rescaled to match the initial provided at $t=0$. That way, it will adjust for the methodological difference between our modelled MtM variability and the model (for example ISDA SIMM) used to actually calculate the IM value.

Based on the simulated MtM paths produced within the IMM batch framework, the detailed methodology for calculating the Dynamic Initial Margin (DIM) at collateral level is outlined in the following steps:

1. Define MtM buckets:

For each time step T_i , the MtM values across all scenarios are sorted and partitioned into M buckets, each containing an equal number of paths. The Dynamic Initial Margin (DIM) is then calculated separately for each bucket.

At present, a single bucket ($M = 1$) is applied. However, a validation process is currently underway to assess the implementation of a more granular bucketing scheme with $M = 20$. Given that 2,000 scenarios are simulated, this configuration would result in 100 scenarios per bucket. This sample size is deemed adequate to support a statistically meaningful estimation of the variance within each bucket.

The use of a single bucket is considered conservative from a risk measurement perspective. A single-bucket approach implies that the offsetting effect arising from the subtraction of the mean MtM variation is applied uniformly across all exposure levels. By contrast, a multi-bucket framework allows this effect to be differentiated according to the MtM level, thereby enhancing risk sensitivity and providing a more granular representation of the DIM profile.

2. Adjustment of MtMs for early maturity trades and forward start trades:

For trades reaching maturity prior to the final simulation horizon, the MtM is set to zero from $T_{maturity}$ onwards, if this date coincides with a simulated time step. Otherwise, the MtM is set to zero from the immediately preceding simulated time step onwards.

For forward-start trades, the MtM is set to zero up to T_{start} , if the start date coincides with a simulated time step. Otherwise, the MtM is set to zero up to the immediately following simulated time step.

3. Computing MtM variations in each bucket and timestep:

For each scenario s in each bucket m , the MtM variation with respect to the previous timestep is calculated as:

$$\Delta MtM_{i-1,i}^{(s,m)} = \sqrt{\frac{MPoR}{T_i - T_{i-1}}} [(MtM_i^{(s,m)} - MtM_{i-1}^{(s,m)}) - \mu_{\Delta MtM}^{(m)}]$$

where $\mu_{\Delta MtM}^{(m)}$ is the mean of $(MtM_i^{(s,m)} - MtM_{i-1}^{(s,m)})$ in the bucket m .

It should be noted that, while IM is conceptually defined as a forward-looking measure, the MtM differences in our model are computed under a backward approximation scheme. Deals maturing between T_{i-1} and T_i are removed from the ΔMtM distribution. Also, the MPoR applied in this calculation will not significantly affect the result due to the re-scaling defined below in point 5.

4. Calculation of VaR as a proxy for IM in each bucket:

Assuming normal behaviour of the MtM variations, the initial margin for bucket m and timestep i is calculated as the VaR at a confidence level $\alpha = 0.99$:

$$\tilde{IM}_i^{(m)} = VaR_{\alpha}(\Delta MtM_{i-1,i}^{(m)}) = N^{-1}(\alpha) StdDev(\Delta MtM_{i-1,i}^{(m)})$$

The initial margin is computed as follows for the internal user and the external counterparty:

$$\tilde{IM}_i^{user(m)} = -N^{-1}(\alpha) StdDev(\Delta MtM_{i-1,i}^{(m)})$$

$$\tilde{IM}_i^{cpty(m)} = N^{-1}(\alpha) StdDev(\Delta MtM_{i-1,i}^{(m)})$$

It should be noted that even though $\alpha = 0.99$ is a typical value for IM calculations, the actual confidence level applied does not significantly affect the result due to the rescaling described in the next point. It should also be noted that for CCR and IMM calculations only the received

initial margin ($\tilde{IM}_i^{cpty(m)}$) is relevant, since the posted initial margin is assumed to be segregated (so that no additional ccr risk arises from posted collateral) from the counterparty in cases where this DIM model is applied.

5. Matching real IM at $t = 0$ and scaling:

The IM calculated through the bucketing approximation will in general differ from the real posted IM. On the other hand, since the Dynamic Initial Margin is based on MtM variations between future timesteps this will take into consideration maturing trades, and gradually decrease as the portfolio matures. However, it could be that the mix of deal types at $t=0$ are different from the ones present at future time steps due to maturing trades, and since we usually expect the initial scaling to match a more conservative (such as ISDA SIMM) methodology than what is implied by MtM variations we will let this initial scaling exponentially decay over time. This means that for future time steps the IM value is given

more and more by the simulated MtM variations, with less weight given to the initial value, which should typically be conservative.

Therefore, a scaling factor is applied to all buckets as follows:

$$IM_i^{(m)} = SF_i \cdot \tilde{IM}_i^{(m)}$$

where

$$SF_i = 1 + (SF_0 - 1) \cdot \exp(-\text{decayFactor} \cdot T_i)$$

$$SF_0 = IM_{\text{official}} / \tilde{IM}_1$$

The IM_{official} is the initial margin calculated by the official calculation system (Calypso) and $\tilde{IM}_1$ value corresponds to the estimated initial margin at T_1 prior to rescaling. There is only one initial margin value at T_1 , since only one bucket can be created at t_0 (as there is only one MtM value at t_0). The IM_{official} corresponds to the amount of IM calculated under ISDA SIMM, without taking into account any IM based on the schedule approach. However, it is important to note that at t_0 the actual held initial margin balance is used to calculate the exposure.

Regarding the exponential scaling, the decay factor chosen is 0.2, which implies a typical time scale of 5 years. However, it should be noted that it is more relevant for XVA calculations than for 1-year exposure. At 1-year, the weighting of the initial scaling is still $e^{-0.2} \approx 82\%$.

6. Final initial margin taking into account collateral agreement threshold:

It should also be noted that the DIM model takes into account characteristics of the IM agreement with the counterparty. In particular, it is typical to have a threshold (for both internal user and external counterparty) for IM calculations below which no IM has to be posted. The threshold amount must be subtracted from the calculated initial margin value before any mitigation of overall exposure is determined. Therefore, the final dynamic initial margin posted by user for each bucket m at the collateral agreement level can be expressed as:

$$DIM_i^{\text{user}(m)} = \min(IM_i^{\text{user}(m)} - \text{threshold}^{\text{user}}, 0)$$

The final dynamic initial margin posted by counterparty for each bucket m at the collateral agreement level can be expressed as:

$$DIM_i^{\text{cpty}(m)} = \max(IM_i^{\text{cpty}(m)} - \text{threshold}^{\text{cpty}}, 0)$$

8.4. Variation Margin Model: Settlement-Adjusted Path Dependant

The Settlement-Adjusted Path-Dependent Model (SAPDM) simulates, **scenario by scenario**, the future variation margin (VM) balances and margin calls under a CSA.

The model follows a **backward-looking approach**: for each reporting date, it reconstructs the VM called up to the start of the margin period of risk (MPoR), and then applies a forward recursive algorithm over the **relevant call dates** to compute margin calls and collateral balances. At the start of the MPoR, the collateral position is determined based on the last margin call date and remains fixed from that point onwards. Its value then evolves according to simulated market risk factors, including FX, which are applied to this position throughout the MPoR horizon. When values are required at dates that are not simulation nodes, the model uses **Brownian-bridge interpolation** to estimate Mark-to-Market (MtM) values and credit-mitigant portfolio MtM to avoid under-capturing MTA effects.

At the initial time t_0 , collateral is represented using the current available mitigant information. When individual mitigant positions are provided, they are modelled through Credit Mitigant Portfolios (CMPs). When such positions are not available, the model uses the aggregated collateral balance and represents it as a synthetic cash mitigant with zero haircut, which is then processed through the same collateral modelling framework.

All calculations related to Variation Margin (MtM, collateral balances and trade flows) are performed in the CSA Agreement Currency. This requires a conversion from the simulation currency into the Agreement Currency using the simulated FX rates, in each scenario and timestep. The resulting collateralised value is then converted into the reporting currency (EUR), using the simulated FX rates again.

8.4.1. Call Dates

The amount of collateral posted and margin call made at a reporting date requires knowledge of margin calls made at previous call dates. Since it is not computationally feasible to compute the margin call at all call dates, the collateral model uses a subset of these.

Then the full set of generated call dates are

$$\Gamma = \{c_{-1}, c_0, c_1, \dots\}$$

determined by the contractual call period p :

$$c_k = t_0 + k \cdot p, \quad k \in \mathbb{N}$$

where c_{-1} is the call date immediately before $c_0 = t_0$, as we need to know how much collateral was called up to t_0 .

The model generates a single set of **relevant call dates**, using the following criteria. For each reporting date r_i , the call dates between $r_i - MPoR$ and $r_i - MPoR + Lag_{VM}$ are considered ($c_{n_i+1}, c_{n_i+2}, \dots, c_{n_i+m_i} \in \Gamma$), where c_{n_i} is the first call date on or before $r_i - MPoR$. Then, to adequately capture the risk arising from MTAs, the amount of VM called on some dates before c_{n_i} are needed (l AdditionalCallDates, $c_{n_i-l}, \dots, c_{n_i-1}$). Consequently, the set of relevant call dates is:

$$\Gamma_{VM}^{(i)} = \{c_{n_i-l}, \dots, c_{n_i-1}, c_{n_i}, c_{n_i+1}, c_{n_i+2}, \dots, c_{n_i+m_i}\}$$

In practice we set **Lag_VM = 0** and **AdditionalCallDates = 1** (that is: no VM lag and one additional pre-call date to capture MTA effects). Moreover, the typical case is that CSAs have **callPeriod = 1** (daily margin calls). In this case, **the relevant call dates set would be reduced to two dates**: the one at the beginning of the MPoR, and one additional call date.

The model uses only the minimum consecutive subset of relevant call dates required to compute the collateral balance and margin call at a given reporting date. Historical call dates beyond that subset are not carried forward, which improves computational efficiency while preserving correctness of the collateral amount at that reporting date. More precisely, for each reporting date r_i , the model does not connect relevant call dates corresponding to different reporting dates. Instead, it considers only the subset of dates required to compute the collateral balance at that reporting date.

8.4.2. MtM at non-simulated call dates (Brownian-bridge)

MtM has to be computed on every call date from c_k to c_{k+n} , being the relevant call dates between two adjacent simulation dates t_i and t_{i+1} . Therefore, for $0 \leq l \leq n$, the MtM on c_{k+l} is computed using Brownian Bridge constructed between c_{k+l-1} and t_{i+1} , where $c_{k-1} = t_i$:

If $l = 0$:

$$MtM^{BB}(c_k, s) = MtM(t_i, s) - TFAdj(t_i, s) + \frac{MtM(t_{i+1}, s) - MtM^{BB}(c_{k+l-1}, s)}{t_{i+1} - t_i} (c_k - t_i) + \sqrt{\frac{(c_k - t_i)(t_{i+1} - c_k)}{(t_{i+1} - t_i)^2}} \varsigma_{k,s}$$

If $0 < l \leq n$:

$$MtM^{BB}(c_{k+l}, s) = MtM^{BB}(c_{k+l-1}, s) + \frac{MtM(t_{i+1}, s) - MtM^{BB}(c_{k+l-1}, s)}{t_{i+1} - c_{k+l-1}} (c_{k+l} - c_{k+l-1}) + \sqrt{\frac{(c_{k+l} - c_{k+l-1})(t_{i+1} - c_{k+l})}{(t_{i+1} - c_{k+l-1})^2}} \varsigma_{k+l,s}$$

where

$TFAdj(t_i, s)$ is the sum of all settlement trade flows occurring between t_i and t_{i+1} , each converted into the agreement currency using the simulated FX rate closest to the flow date (if a flow date is exactly equidistant between two simulation nodes, the FX rate of the late simulation node is used)

$\varsigma_{i,s}$ is a random normal draw with zero mean and covariance matrix VCV_{cmp} (see [Variance Covariance \(VCV\) Matrix](#) for its calculation).

Then, the simulated MtM is:

$$MtM(c_{k+l}, s) = MtM^{BB}(c_{k+l}, s) + \sum_{c_{k+l} \leq d \leq t_{i+1}} TF(d, s)$$

where $TF(d, s)$ is the sum of trade flows converted into the agreement currency at the settlement date d .

Counterparty/user credit mitigation portfolio MtM after applying relevant contractual collateral haircuts for each CMP_A :

If $l = 0$:

$$CMtMHC_A^{BB}(c_k, s) = CMtMHC_A(t_i, s) + \frac{CMtMHC_A(t_{i+1}, s) - CMtMHC_A(t_i, s)}{t_{i+1} - t_i} (c_k - t_i) + \sqrt{\frac{(c_k - t_i)(t_{i+1} - c_k)}{(t_{i+1} - t_i)^2}} \varsigma_{k,s}$$

If $0 < l \leq n$:

$$CMtMHC_A^{BB}(c_{k+l}, s) = CMtMHC_A^{BB}(c_{k+l-1}, s) + \frac{CMtMHC_A(t_{i+1}, s) - CMtMHC_A^{BB}(c_{k+l-1}, s)}{t_{i+1} - c_{k+l-1}} (c_{k+l} - c_{k+l-1}) + \sqrt{\frac{(c_{k+l} - c_{k+l-1})(t_{i+1} - c_{k+l})}{(t_{i+1} - c_{k+l-1})^2}} \varsigma_{k+l,s}$$

$$CMtMHC_{cpty}(c_{k+l}, s) = \max(CMtMHC_{cpty}^{BB}(c_{k+l}, s), 0)$$

$$CMtMHC_{user}(c_{k+l}, s) = \min(CMtMHC_{user}^{BB}(c_{k+l}, s), 0)$$

8.4.2.1. Variance Covariance (VCV) Matrix

On a given non-simulated call date, maximum three MtM values need to be computed:

1. Trade portfolio MtM

2. $CMtMHC_{cpty}$

3. $CMtMHC_{user}$

Let's define:

$$x_{i,s} = MtM(t_{i+1}, s) - (MtM(t_i, s) - TFAdj(t_i, s))$$

$$y_{i,s} = CMtMHC_{cpty}(t_{i+1}, s) - CMtMHC_{cpty}(t_i, s)$$

$$z_{i,s} = CMtMHC_{user}(t_{i+1}, s) - CMtMHC_{user}(t_i, s)$$

The estimation of 3x3 VCV matrix, Σ , used to compute MtM values on non-simulated call dates that are between adjacent simulation dates t_i and t_{i+1} , is calculated this way:

$$V_p(t_i) = \frac{1}{N-1} \sum_{j=1}^N (x_{i,s_j} - \bar{x}_i)^2$$

$$V_{cmp_cpty}(t_i) = \frac{1}{N-1} \sum_{j=1}^N (y_{i,s_j} - \bar{y}_i)^2$$

$$V_{cmp_user}(t_i) = \frac{1}{N-1} \sum_{j=1}^N (z_{i,s_j} - \bar{z}_i)^2$$

$$COV_{p, cmp_cpty}(t_i) = \frac{1}{N-1} \sum_{j=1}^N (x_{i,s_j} - \bar{x}_i)(y_{i,s_j} - \bar{y}_i)$$

$$COV_{p, cmp_user}(t_i) = \frac{1}{N-1} \sum_{j=1}^N (x_{i,s_j} - \bar{x}_i)(z_{i,s_j} - \bar{z}_i)$$

$$COV_{cmp_cpty, cmp_user}(t_i) = \frac{1}{N-1} \sum_{j=1}^N (y_{i,s_j} - \bar{y}_i)(z_{i,s_j} - \bar{z}_i)$$

where

$$\bar{x}_i = \frac{1}{N} \sum_{j=1}^N x_{i,s_j}, \bar{y}_i = \frac{1}{N} \sum_{j=1}^N y_{i,s_j}, \bar{z}_i = \frac{1}{N} \sum_{j=1}^N z_{i,s_j}$$

Then the random normal vector $\varsigma \sim N(0, \Sigma)$ used in Brownian Bridge computation is created as:

$$\zeta = zA^T$$

where

z is a 1×3 vector of independent random numbers following the standard normal distribution, and A is the linear transformation matrix obtained via a stable version of eigenvalue decomposition of VCV matrix such that $\Sigma = AA^T$.

There is a separate random draw for each MtM^{BB} , $CMtMHC_{cpty}^{BB}$ and $CMtMHC_{user}^{BB}$ computation.

8.4.3. Initial Variation Margin Balance (IVMB)

The model first determines the **Initial Variation Margin Balance (IVMB)** at t_0 , which is the variation margin available at the start of the simulation (up to and including c_{-1}).

$$IVMB = \sum_{m=1}^M CMtM_m^{sys}(t_0)$$

where

$CMtM_m(t_0)$ is the source system MtM in agreement currency for mitigant m , and M is the total number of mitigants.²

When **physical collateral** is modelled, the engine does not attempt to track every individual pledged asset continuously. Instead, it represents the pledged pool on each side of the agreement as a small number of representative portfolios. Each representative portfolio, **CMP (Credit-Mitigant Portfolio)**, aggregates the actual mitigants that the party may post under the collateral agreement and therefore acts as a single model object that captures the composition, market value and liquidity characteristics of the physical collateral set.

A single numeric factor — the **position unit (PU)** — is associated with each representative portfolio. Conceptually the position unit is a scale factor that converts the portfolio's relative composition into the absolute quantity of collateral that the model assumes is available or required. The position unit is fixed so that, at the valuation start, the scaled representative-portfolio value matches the actual collateral balance observed in the books. Thereafter, **the composition of mitigants remains constant through time.**

Creation of the CMPs:

² For STM trades, the IVMB is obtained directly from the simulated MtM values rather than computed through the standard margin call algorithm, as daily settlement effectively resets the mark-to-market.

At t_0 , the engine constructs two representative collateral portfolios. One portfolio aggregates all the assets that the counterparty is expected to deliver to the bank, CMP_{cpty} , the other aggregates the assets the bank is expected to deliver to the counterparty, CMP_{user} .

Compute the position unit (PU) of CMP at t_0 :

$$CMtM_A^{sim}(t_0) = \sum_{m=1}^M PU_{A,m} * CMtM_{A,m}^{sim}(t_0)$$

where

$A \in \{cpty, user\}$

$PU_{A,m}$: position unit of mitigant m at t_0

$CMtM_{A,m}^{sim}(t_0)$: simulated MtM of mitigant m in CMP_A at t_0

M : total number of mitigants in CMP_A .

Then, the position unit of CMP_A at t_0 is computed this way:

$$PU_A(t_0) = \begin{cases} \max\left(\frac{IVMB}{CMtM_A^{sim}(t_0)}, 0\right), & \text{if } CMtM_A^{sim}(t_0) \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

In case $CMtM_A^{sim}(t_0) = 0$, the future collateral posted by A is assumed to be in cash.

Note that there are two different uses of the term position unit. $PU_{A,m}$ refers to the weight of each mitigant within the CMP, while $PU_A(t)$ refers to the number of units of the originally defined CMP, which will later vary with time.

Then, at each call date c and under scenario s , obtain the trade portfolio MtM ($MtM(c, s)$) and the credit mitigation portfolio MtM after applying all relevant contractual collateral haircuts ($CMtMHC_A(c, s)$) for each CMP_A . These haircuts reflect that when securities are posted as collateral an additional amount is typically required to cover the same exposure, to account for price volatility in these securities. These haircuts will increase both the amount of received and posted securities.

8.4.4. Variation Margin Required (VMR) at a call date

The amount of **variation margin required** under scenario s is defined as:

$$X(c, s) = MtM(c, s) + CPIIndependent + UserIndependent$$

$$VMR_{NoCap}(c, s) = \begin{cases} X(c, s) - cVMT(c), & \text{if } X(c, s) > cVMT(c) \text{ (i.e. if cpty posts to bank)} \\ X(c, s) - uVMT(c), & \text{if } X(c, s) < uVMT(c) \text{ (i.e. if bank posts to cpty)} \\ 0, & \text{otherwise} \end{cases}$$

where

$cVMT$: counterparty variation margin threshold

$uVMT$: user variation margin threshold

$CPIIndependent$ ($UserIndependent$) is the CSA counterparty (user) independent amount.

Trade-level independent amounts are currently not included in the VMR calculation and are instead applied later in the exposure calculation through initial profiles (see [Collateralised exposure](#)).

Note: Independent Amounts are converted into the CSA Agreement currency using the FX rates simulated at the nearest scenario grid node. If a date is exactly equidistant between two simulated nodes, the FX rate from the later simulated date is used.

If a variation margin cap is present in the collateral agreement:

$$VMR(c, s) = \begin{cases} \min(VMR_{NoCap}(c, s), cVMCap(c)), & \text{if } VMR_{NoCap}(c, s) \geq 0 \\ \max(VMR_{NoCap}(c, s), uVMCap(c)), & \text{if } VMR_{NoCap}(c, s) < 0 \end{cases}$$

where

$cVMCap$: counterparty variation margin cap

$uVMCap$: user variation margin cap.

8.4.5. Margin call (MC) and recursive update (algorithm)

Within the recursive algorithm, collateral can be represented either through Credit Mitigant Portfolios (CMP), whose position units are updated using margin calls, or through cash collateral balances when no CMP is available. Both representations are updated consistently using the same margin call mechanism.

- At c_j , the position unit of each CMP_A are computed as:

$$PU_A(c_j, s) = \begin{cases} PU_A(t_0), & \text{if } c_j = c_{-1} \\ \max(\frac{VMR(c_j, s)}{CMtMHC_A(c_j, s)}, 0), & c_j \neq c_{-1} \end{cases}$$

The amount of cash collateral (when no CMP is available) posted by counterparty is computed as:

$$CashCollateral_{cpty}(c_j, s) = \begin{cases} \max(IVMB, 0), & \text{if } c_j = c_{-1} \\ \max(VMR(c_j, s), 0), & \text{if } c_j \neq c_{-1} \end{cases}$$

- At c_{j+l} , for $1 \leq l \leq k$:

$$Y(c, s) = VMR(c_{j+l}, s) - \sum_A (PU_A(c_{j+l-1}, s) * CMtMHC_A(c_{j+l}, s) + CashCollateral_A(c_{j+l-1}, s))$$

$$MC_{user}(c_{j+l}, s) = \begin{cases} 0, & \text{if } uVMMTA(c) < Y(c, s) < cVMMTA(c) \\ \text{round}[Y(c, s), RA], & \text{otherwise} \end{cases}$$

where

function $\text{round}(x, m)$ rounds x to the nearest integer multiple of m

$cVMMTA$ is the counterparty Variation Margin Minimum Transfer Amount

$uVMMTA$ is the user Variation Margin Minimum Transfer Amount

RA is the round-off amount

The **total position unit** of CMP_{cpty} is computed as:

$$PU_{cpty}(c_{j+l}, s) = \max[PU_{cpty}(c_{j+l-1}, s) + \frac{MC(c_{j+l}, s) + PU_{user}(c_{j+l-1}, s) * CMtMHC_{user}(c_{j+l}, s) + CashCollateral_{user}(c_{j+l-1}, s)}{CMtMHC_{cpty}(c_{j+l}, s)}, 0]$$

(equivalently for $PU_{user}(c_{j+l}, s)$)

The **incremental position unit** of CMP_A is computed as:

$$\Delta PU_A(c_{j+l}, s) = PU_A(c_{j+l}, s) - PU_A(c_{j+l-1}, s)$$

When a CMP is unavailable, the cash collateral balances reflect the cumulative effect of margin calls over time and are updated consistently within the same recursive mechanism used for CMP position units. Their evolution captures the net collateral position between both parties and is computed using information from both sides of the agreement, taking into account the total collateral position and its allocation between the user and the counterparty, thus reflecting the bilateral nature of the margining process.

The **total amount of cash collateral** posted by user is computed as:

$$\begin{aligned} CashCollateral_{user}(c_{j+l}, s) = & \min[\Sigma_A CashCollateral_A(c_{j+l-1}, s) \\ & + MC(c_{j+l}, s) + PU_{cpty}(c_{j+l-1}, s) * CMtMHC_{cpty}(c_{j+l}, s), 0] \end{aligned}$$

(equivalently $CashCollateral_{cpty}(c_{j+l}, s)$)

The **incremental amount of cash collateral** posted by A is computed as:

$$\Delta CashCollateral_A(c_{j+l}, s) = CashCollateral_A(c_{j+l}, s) - CashCollateral_A(c_{j+l-1}, s)$$

8.4.6. Variation Margin Smooth and Variation Margin Full

The **Variation Margin balance** used for the calculation of the **smooth exposures** is defined as:

$$VM_{smooth}(r_i, s) = \Sigma_A [PU_A(c_{n_i}, s) * CMtM_{A,default}(r_i, s) + CashCollateral_A(c_{n_i}, s)]$$

where $CMtM_{A,default}(r_i, s)$ is the MtM of CMP_A after applying **haircuts at default**.

The **Variation Margin balance** used for the calculation of the **full exposures** is defined as:

$$VM_{full}(r_i, s) = \Sigma_A [PU_A(c_{n_i+m_i}, s) * CMtM_{A,default}(r_i, s) + CashCollateral_A(c_{n_i+m_i}, s)]$$

where

$$PU_A(c_{n_i+m_i}, s) = PU_A(c_{n_i}, s) + \left\{ \begin{array}{l} \sum_{l=1}^{m_i} \frac{\min(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} * \Delta PU_A(c_{n_i+l}, s) \text{ from user's perspective} \\ \sum_{l=1}^{m_i} \frac{\max(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} * \Delta PU_A(c_{n_i+l}, s) \text{ from counterparty's perspective} \end{array} \right.$$

$$CashCollateral_A(c_{n_i+m_i}, s) = CashCollateral_A(c_{n_i}, s) +$$

$$\left\{ \begin{array}{l} \sum_{l=1}^{m_i} \frac{\min(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} * \Delta CashCollateral_A(c_{n_i+l}, s) \text{ from user's perspective} \\ \sum_{l=1}^{m_i} \frac{\max(0, MC(c_{n_i+l}, s))}{MC(c_{n_i+l}, s)} * \Delta CashCollateral_A(c_{n_i+l}, s) \text{ from counterparty's perspective} \end{array} \right.$$

8.4.7. Collateralised exposure

The **expected collateralized exposure**, at time t_j for a counterparty CP , is defined as:

$$\begin{aligned} EE(t_j) &= \max[\sum_{MA \in CP} (\frac{1}{N} \sum_{k=1}^N \max(CV_{MA}(t_j, s_k), 0) + IP_{MA}(t_j)), 0] \\ &= \max[\sum_{MA \in CP} (\frac{1}{N} \sum_{k=1}^N \max(\sum_{CSA \in MA} CV_{CSA}(t_j, s_k), 0) + IP_{MA}(t_j)), 0] \end{aligned}$$

where

$IP(t)$ denotes the time-dependent value of initial profiles, which incorporate trade-related independent amounts and reflect the maturity of the underlying positions

$CV_{CSA}(t_j, s_k)$:

- smooth case:

$$\begin{aligned} CV_{CSA}(t_j, s_k) &= MtM_{CSA}(t_j, s_k) - DIM_{cpty}(t_j, s_k) - VM_{CSA}^{smooth}(t_j, s_k) \\ &+ NetTF_{CSA}^{MPoR}(t_j, s_k) - IA_{cpty}^{deal}(t_j, s_k) \end{aligned}$$

- full case:

$$CV_{CSA}(t_j, s_k) = MtM_{CSA}(t_j, s_k) - DIM_{cpty}(t_j, s_k) - VM_{CSA}^{full}(t_j, s_k) \\ + NetTF_{CSA}^{MPoR}(t_j, s_k) - TF_{CSA}^{Lag_{TS}}(t_j, s_k) - IA_{cpty}^{deal}(t_j, s_k)$$

where

$MtM_{CSA}(t_j, s_k)$: aggregated MtM of all the trades in the CSA

$NetTF_k^{MPoR}(t, s)$: the cash flows of transactions during the MPOR

$TF_k^{Lag_{TS}}(t, s)$: the total trade settlement flow paid by the non-defaulting party during the Lag_{TS}

$IA_{cpty}^{deal}(t_j, s_k)$: aggregated Independent Amounts of all the trades in the CSA, converted to the CSA currency

Note: Independent Amounts are converted into the CSA Agreement currency using the FX rates simulated at the nearest scenario grid node. If a date is exactly equidistant between two simulated nodes, the FX rate from the later simulated date is used.

It should be noted that all **trade-related cash flows occurring on the same date are netted prior to their inclusion in the exposure calculation, regardless of their currency.**

It should also be noted that the **aggregation** of the collateralised value follows the legal hierarchy of the collateral agreements. First, the collateralised value is calculated at **CSA level**. The resulting values are then aggregated across all CSAs belonging to the same **Master Agreement** (MA). Finally, the values obtained for each MA are aggregated at **counterparty level**, and the collateralised exposure used in the exposure profile is defined as the positive part of this aggregation.

In practice, the general approach within BBVA is to execute a single Credit Support Deed (CSD) per Master Agreement, typically covering trades under different CSAs within that Master Agreement. Consequently, although the model is capable of computing Dynamic Initial Margin at CSA level, in the portfolios currently in scope DIM is effectively aggregated and mitigated at Master Agreement level. In the few cases where more than one IM agreement exists within the same Master Agreement, the methodological treatment would remain unchanged.

8.4.8. Settled-to-Market convention

Certain collateral agreements –typically associated with swap agent structures, organised markets or central clearing counterparties (CCPs)– follow a **Settled-to-Market (STM)** convention. Under

these agreements, the mark-to-market (MtM) of transactions is settled on a daily basis, so that gains and losses are exchanged continuously and the MtM is reset to zero after each settlement.

From a modelling perspective, these agreements are represented through the activation of the SettledToMarketAgreement flag at CSA level, which leads to the following characteristics:

- Full collateralisation with no thresholds or minimum transfer amounts, i.e.:
 - Threshold = 0
 - MTA = 0
- No independent amounts, i.e.:
 - CPindependent = 0
 - UserIndependent = 0
- Daily margining frequency, with:
 - CallPeriod = 1
- The initial variation margin balance (IVMB) is directly derived from the portfolio MtM at the valuation date, reflecting the fact that the MtM is continuously settled.

As a result, exposures under STM agreements are typically close to zero outside the MPoR, and the residual risk is mainly driven by short-term market movements between settlement dates.

This treatment ensures that the model adequately reflects the economic reality of daily-settled transactions and avoids overstating exposures for centrally cleared or equivalent structures.

8.5. Spikes and Smooth Exposure

To correctly consider the modelling of the spikes, two exposure calculations are performed.

In one calculation it is assumed that neither BBVA nor the non-defaulting party makes margin or trade settlement flow payments after the beginning of the MPOR. The resulting exposure profile is the **smooth exposure**, where the only movement considered during the MPoR is that of the MtM.

In the other calculation, it is assumed that the defaulting party ceases margin and trade settlement payments after the beginning of the MPOR, while BBVA continues to make variation margin and trade settlement payments according to its default management process (see [8.5. Default Management Process](#)). This assumption reflects the asymmetry in post-default behavior and produces the **full exposure** profile, which typically shows discontinuities or jumps due to the mismatch in cashflows.

The difference between both exposure profiles represents the **spikes**. They represent the impact of unilateral margin and settlement flows during the MPoR, which are only made by the non-defaulting party. These spikes reflect the temporary over- or under-collateralization that can occur due to the non-reciprocal behavior in the event of counterparty default.

Then, the **settlement spike** caused by non-defaulting party's margin and trade settlement flow payments after the beginning of the MPoR is computed as:

$$SettlementSpike(t, s) = Exp_{full}(t, s) - Exp_{smooth}(t, s)$$

To properly capture the impact of these spikes in capital calculations, an exposure add-on is computed (see [12. RNIEPEs](#)).

8.5.1. CLS

Trades settled through CLS (**Continuous Linked Settlement**) use a Payment versus Payment (PvP) mechanism to ensure that both sides of a foreign exchange transaction are settled simultaneously. Therefore, those trades eliminate settlement risk and tend to experience significantly fewer spikes compared to trades not settled via CLS.

8.5.2. Swap Agent

In contrast, trades processed through a Swap Agent (e.g. LCH SwapAgent) may still involve settlement risk. Swap Agents are not settlement systems and do not guarantee PvP by default. Whether PvP is achieved depends on the specific agreement and infrastructure used by the counterparties. As a result, trades settled via Swap Agent may show more frequent or pronounced spikes, particularly around payment dates.

8.6. Default Management Process

The Default Management Process (DMP) refers to all legal and operational actions undertaken by the institution following a counterparty default, prior to ceasing margin calls and trade-related cash flow payments to the defaulted counterparty.

In the absence of a well-defined DMP, all contractual cash flows owed by the institution are assumed to continue throughout the full Margin Period of Risk (MPoR), as indicated in the *ECB Guide to Internal Models – Counterparty Credit Risk, paragraph 3.2.20*. The DMP defined in BBVA enables the bank to detect a counterparty default as early as possible and to stop posting collateral before the end of the MPoR. As a result, BBVA can adopt a more proactive approach when simulating the impact of future defaults on exposures, and therefore on counterparty credit risk capital.

The DMP exercise takes an average of two days to compile and analyze the counterparty's situation by the various CIB departments involved: front office, back office, risk, client analysts, and legal. At the end of this period, the counterparty would be notified of the failure to pay, which would enable

the entity to stop the outflow of cash at the end of the grace period. Cash flows in the model takes into account this reality when computing the spikes during MPoR.

8.7. Penalties in the MPoR

Just as the DMP allows the MPoR to be shortened, there are several conditions under which it must be extended:

- For all netting sets where the number of transactions exceeds 5,000 at any time during a quarter, the margin risk period in the following quarter shall be at least 20 business days (Article 285(3.a) CRR).
- Where netting sets include illiquid collateral or OTC derivatives that cannot be readily substituted under stressed market conditions, the MPoR shall also be at least 20 business days (Article 285(3.b) CRR).
- If an institution has been involved in more than two margin call disputes for a given netting set over the two preceding quarters, and those disputes have lasted longer than the MPoR as defined in the previous two cases, the institution must apply an MPoR at least twice as long as that specified above for the affected netting set during the subsequent two quarters (Article 285(4) CRR).

For more details:

[IMM - MPOR - Penalizaciones Colaterales](#)

8.8. Haircuts

A **contractual collateral haircut** is applied based on the type, credit quality, and liquidity of the asset being posted as collateral. Generally, the higher the quality and liquidity, or the stronger the creditworthiness of the issuer, the lower the contractual haircut negotiated in the collateral agreement.

These haircuts are applied to physical collateral and increase its effective value for margining purposes, as they result in a larger amount of collateral being posted or received to cover a given exposure.

The **Haircut at Default (HaD)** represents an additional discount applied to the value of physical collateral in the event of default, in order to reflect the potential losses under stressed liquidation conditions.

Within the IMM framework, both contractual haircuts and haircuts at default are applied to the MtM of the collateral, when the collateral consists of securities, by scaling its value. For contractual haircuts, an additional amount of collateral is posted or received, in both directions (i.e. for both

posted and received collateral), according to the haircuts specified in the collateral agreement for the relevant type of collateral.

$$\text{Adjusted Value Contractual Haircut} = MtM * (1 + \text{Haircut})$$

Similarly, for HaD, a regulatory specified haircut is applied, asymmetrically according to whether the collateral is posted or received.

$$\text{Value of posted collateral} = \text{Posted Security Collateral} * (1 + \text{Haircut})$$

$$\text{Value of received collateral} = \text{Received Security Collateral} * (1 - \text{Haircut})$$

A **FX haircut** may apply under the SA-CCR framework when there is a currency mismatch between the collateral and the CSA Agreement Currency, as defined in Article 224 of the CRR.

However, within the IMM framework, no explicit FX haircut is applied, as foreign exchange risk is captured through the dynamic simulation of FX rates.

More details can be found in this document:

[C102 - SAPDM PhysicalCollateralCashMitigants+HCs WholePortfolio](#)

9. Backtesting

9.1. Risk Factors Backtesting

The methodology basically consists of simulating risk factor paths during the backtesting period and, based on these, defining an upper and lower bound (envelope) that will serve as a reference to calculate, for both each simulated path and the realized factor path, the percentage of time (tau) that each path is outside that envelope. These values are scaled by time, to account for the fact that the grid is not daily. The lower this calculation is for the actual path compared to the tau distribution of the simulated paths, the better the backtesting results. In this way, we define bands to determine whether the model is correct (green), incorrect (red), or inconclusive (yellow).

Backtesting is also performed based on the distance to the envelope.

The list of risk factors selected for backtesting is based on a ranking by sensitivity.

All the details are explained in this document:

[Backtesting of Risk Factors IMM BBVA - Methodology](#)

The regulation requires backtesting across different time horizons (Article 294.1(a) CRR). To this end, the aforementioned methodology is applied to shocks with different horizons: 7, 14, 28, and 91 days. More details can be found in this document:

[RF Backtesting v1.0 1](#)

9.2. Portfolio Backtesting

Similar to risk factor backtesting, this methodology involves simulating exposure paths during the backtesting period and defining an upper and lower bound (envelope) from them. This will serve as a reference for calculating, for both each simulated path and the actual exposure path, the percentage of time (τ) that each path is outside that envelope. These values are scaled by time, to account for the fact that the grid is not daily. The lower this calculation is for the actual path compared to the τ distribution of the simulated paths, the better the backtesting results. Again, we define bands to determine whether the model is correct (green), incorrect (red), or inconclusive (yellow).

Backtesting is also performed based on the distance to the envelope.

All the details are explained in this document:

[Backtesting of Portfolio Exposure IMM BBVA - Methodology](#)

As suggested in Article 294 CRR, this backtesting methodology is applied to the following portfolios:

- Real portfolio:

The actual portfolio is the set of instruments existing on the start date of the backtesting process. It is considered semi-static during the backtesting period, as no additional instruments are added during this period, but the maturities of existing instruments are respected if they occur during this period.

The distribution of collateral and variation margin attributes is performed in the same manner as in the general exposure modeling. However, for the purpose of backtesting, exposures of the real portfolio are computed **excluding initial margin**.

- Representative portfolio:

A series of portfolios has been defined, consisting of a set of transactions representative of the bank's portfolio and a fictitious set of collateral contracts representing different exposure mitigations (bilateral contracts with different MPoR lengths, to represent different backtesting horizons applying a perfect CSA). This includes a set of portfolios consisting of a single trade, for the most relevant

trade types. Backtesting is performed on each of these portfolios combined with each proposed collateral case study. More details can be found in this document:

[C102 - Backtesting v1.0](#)

9.3. Global Backtesting, traffic lights

The backtesting methodology explained in the previous sections provides individual results for each counterparty or each risk factor in terms of the percentage of time out of the envelope or in terms of the distance to the envelope. In order to establish a global result for each backtesting process an aggregation methodology has been implemented. Based on the correlation matrix of the realized paths, the cumulative density function (CDF) of the ratio of breaches is calculated via the simulation of a N-dimensional copula. This simulation is performed assuming that all the counterparties or risk factors have the same probability to cross the envelope. Once the CDF is calculated, the different thresholds (red, amber and green) of the global backtesting are calculated in terms of the ratio of breaches. From the individual results, the realized ratio of breaches is calculated and compared to the thresholds in order to determine the final color for each backtesting type. In the case of risk factors an independent result is given for each horizon.

More details can be found in this document:

[IMM_Global_Backtesting.pdf](#)

10. Stress Testing

The stress testing process has the purpose to analyze different scenarios that could affect the viability of the institution, as well as serving as an input for the adequacy of the current stressed period used for the Historical Stress (HS) calibration.

The precise methodological definition of how the stress testing calculations are performed is described in [BBVA Exposure Stress Testing](#), so we will only provide an overview here.

In general, we have defined three types of stress testing:

1. Stressed Calibration, in which we test different historical periods for calibrating the exposure model. This type of stress will provide input on the adequacy of the current stressed period for Historical Stress calibration.
2. Stressed Initial Conditions, in which the current market state is stressed. This type of stress can be used to assess capital adequacy. In these scenarios, for the collateralized counterparties a perfect collateralization is considered at t_0 (ModelledVMt0=True).

3. Reverse Stress Test, in which we ascertain a given movement in market variables which would correspond to a particular unfavorable outcome. In this case we are checking what would cause a breach in the current limits.

10.1. Extra Stressed Period Search

In the case of the extra stressed period search we are taking a page from FRTB IMAs playbook in the sense that we want to search for all possible stressed calibration windows in order to check whether the result is consistent with our current HS window.

We want to scan all possible one-year windows. This should provide a result more stressed than the three-year calibration underlying HS, since the worst one-year period will always be more stressed than any given three-year period.

However, it is not possible to perform an exact exposure calculation for every possible calibration window in the case of counterparty credit risk, due to orders of magnitude more calculations being necessary (For market risk, moving the window forward one day only involves a small set of portfolio valuations, while a CCR exposure calculation entails hundreds of thousands of such valuations). So in order to perform this search we will perform more of a market risk type search, in which we maximize the portfolio volatility based on MtM sensitivities. However, we will align the result with CCR in the sense that we will mitigate the effect of sensitivities corresponding to collateralized counterparties. This is important for CCR because even if the MtM sensitivity is close to zero in market risk terms (due to P&L management), it might not be so for CCR if an uncollateralized exposure is hedged by a collateralized exposure.

For this reason we will use available information, which in our case is a reduced set of CVA sensitivities, to estimate how collateralized sensitivities should be reduced due to collateralization. This is done by calculating the ratio of CVA to MtM sensitivities across all collateralized, and then across all non-collateralized counterparties. These ratios can be used to scale each type of sensitivity.

After performing this “CCR scaling” of all sensitivities we can then estimate the portfolio standard deviation based on correlations and volatilities calibrated to a given historical one-year period, and then scan through all possible such one year periods in order to maximize this volatility.

After the resulting one-year period has been determined a complete exposure calculation is calibrated and calculated in a complete CCR batch. This provides stressed exposures corresponding to “extra stress”. By construction this should provide higher exposures than the standard HS calculation. Furthermore, if the resulting window falls outside of the HS window this is relevant for considering whether this window should be recalibrated.

10.2. Lehman and Covid

Apart from the Extra stressed period search we also calibrate and calculate CCR batches based on two specific historical periods, the Lehman period (defined as 2008, in accordance with the current stress scenario of the same name in our current internal market risk model), and the Covid period, defined as the year 2020.

These calculations serve as information on how the model calibration would behave if market conditions similar to these rather singular historical periods would return, as well as providing a challenge to the Extra Stressed Period Search methodology, since the latter should always provide a more conservative result since it scans through all possible stress periods.

10.3. Forward Looking Implicit

We now move on to stress tests in which we change the current market variables, the so-called initial conditions. These stress tests are called “Forward Looking” since we are looking forward to what could happen in the near future, in order to anticipate adverse possibilities.

In the case of the forward looking stress test we will use market implied volatilities (which reflects the markets expectation on what could happen in the upcoming months) in order to generate future scenarios at a given time horizon. We will then apply the same (CCR scaled) sensitivities that are used for the Extra Stressed Period Search in order to estimate the portfolio impact in these scenarios.

A scenario is then selected based on a given confidence level, and a complete CCR batch is run after having impacted the initial conditions with the impacts in this scenario.

The Implicit forward looking should provide an exposure increase by construction since the scenario is selected based on portfolio sensitivities.

10.4. Forward Looking ICAAP

A second stress test which impacts initial conditions in a forward looking manner is the ICAAP stress test. This stress test is based on the impacts provided in the yearly bankwide ICAAP exercise, which are kept constant during the entire year, but then extrapolated in each stress test execution to a more complete set of risk factors relevant for the CCR portfolio at each execution date.

ICAAP exercise provides impacts to a reduced set of risk factors for each typology. An independent generalization of impacts is performed for each typology. For equity and forex risk factors, the extrapolation of the impacts to the unknown risk factors is carried out considering that all risk factors follow a multivariate gaussian with parameters based on historical data and correlated with the risk factors with known impacts. In the case of IR risk factors, the generalization of impacts is performed in two steps. ICAAP exercise usually provides shocks for a reduced number of pillars of each specific curve. Therefore, an impact extrapolation to all pillars of the curves of a specific currency is first performed based on a principal component analysis (PCA). Afterwards, a generalization to the whole

set of IR curves is performed. This second step is performed considering that the two first principal components of each curve with unknown impacts follow a multivariate gaussian correlated with the two first principal components of the curves with known impacts (the curves at the previous step). All the parameters of the multivariate gaussian are again based on historical data.

After having impacted the initial conditions by applying the ICAAP shocks, a complete CCR batch is run, producing stressed exposures.

10.5. Reverse Stress Testing

A reverse stress test consists in determining the conditions which would lead to a particular unfavorable outcome.

In this context, we will take exposure increases which would breach the current CCR exposure limits as the relevant unfavorable outcome.

In order to find this scenario we will take the Forward Looking Implicit scenario as a starting point, since it will lead to an adverse scenario based on the composition of our portfolio. CCR exposures will be estimated under this Forward Looking scenario, capturing relevant aspects, such as collateralization and non-linearity of the portfolio. We then compare the resulting exposure increase with what would breach the CCR exposure limit, and rescale the impacts in the scenario accordingly. The output of this analysis is a set of shocks in market variables which correspond to this adverse outcome.

11. Wrong Way Risk

11.1. General WWR

General Wrong-Way Risk arises when the likelihood of default by counterparties is positively correlated with general market risk factors. In other words, the exposure to a counterparty tends to increase in scenarios where systemic market movements—such as falling equity indices, interest rate shocks, or FX depreciation—also lead to a deterioration in the counterparty's creditworthiness. This correlation implies that the institution may face higher exposures precisely when counterparties are more likely to default, amplifying the risk. gWWR is particularly challenging to model consistently, as it depends on dynamic and often non-linear relationships across market conditions and counterparties.

The IMM acknowledges the difficulty in directly modelling general wrong-way risk. As a result, institutions are not required to explicitly quantify or simulate gWWR in their IMM exposure profiles.

Instead, the regulation provides a pragmatic alternative through the application of a regulatory multiplier: the alpha factor.

The alpha factor, set at 1.4 by regulatory standards, serves as a buffer or multiplier within the exposure-at-default (EAD) calculation. This factor is intended to compensate for model risk, approximation errors, and unmodelled effects such as gWWR.

11.1.1. Methodological Framework

11.1.1.1. Credit State Evolution in RiskWatch

BBVA uses RiskWatch (Algorithmics) to simulate the credit state evolution of counterparties. The model incorporates both:

- A **Credit Worthiness Process (CWP)** driven by market and idiosyncratic risk factors.
- A **Credit State Process (CSP)** governed by a transition matrix calibrated from historical data.

This hybrid approach allows the model to capture the evolution of credit risk under changing market scenarios while maintaining probabilistic integrity with historical data.

11.1.1.2. Systemic Credit Scalars

The systemic drivers of creditworthiness are modelled as **Brownian motion processes**, calibrated from historical **5-year CDS spreads**. These drivers reflect macroeconomic conditions impacting broad groups of counterparties. The **granularity** of these factors is a key modelling decision.

BBVA evaluated three levels:

- **EurScalar (General)**: A single systemic factor.
- **Geography**: Systemic factors per geographic region.
- **GeoSecRating**: Most granular, including geography, sector, and rating.

11.1.1.3. Simulation and Calibration

The systemic credit scalars are simulated jointly with other risk factors (interest rates, FX, etc.), preserving historical correlations. This integrated approach ensures that exposure simulations reflect realistic interactions between market variables and creditworthiness, which is essential for capturing gWWR.

The simulation process is calibrated using:

- A **variance-covariance (VCV) matrix** of systemic credit scalars.
- **Proxy CDS**, ensuring systemic risks are captured without excessive noise from idiosyncratic variation.

11.1.2. Implementation and Practical Considerations

11.1.2.1. Specific Weights and IRC Model Integration

The idiosyncratic component of credit volatility is taken from the **IRC (Incremental Risk Charge) model**, which uses a bifactorial credit risk model consistent with regulatory approvals for market risk. This ensures internal consistency across models and leverages already validated inputs.

11.1.2.2. Granularity Decision

Although finer granularity theoretically improves precision, it increases computational burden and sensitivity to noisy inputs. BBVA's empirical analysis found:

- Geography-level granularity provides a sufficiently conservative and stable measure of systemic risk.
- Differences in exposure outcomes between Geography and GeoSecRating were minimal (<5%).
- EurScalar showed materially different outcomes and was not reliably conservative under all portfolio conditions.

As a result, **Geography-based granularity** was selected as the most appropriate balance between model accuracy and robustness.

11.1.2.3. Exposure Assumption: Smooth vs Full

For consistency with capital calculations, BBVA uses **Smooth Exposure** in gWWR modelling.

11.1.3. Monitoring and Governance

The model is executed regularly and monitored to ensure that Expected Positive Exposure (EPE), including gWWR, does not exceed the capital buffer implied by the alpha factor. Breaches may trigger supervisory review or lead to additional AddOns.

Importantly, BBVA does not intend to apply for an internal model to reduce the alpha factor. The regulatory alpha of 1.4 is maintained, and the focus is on ensuring that the internal simulation framework supports its adequacy under plausible adverse conditions.

For more information see the [gWWR Scenario Generation Methodology Document](#).

11.2. Specific WWR

SWWR arises when a counterparty's default risk is positively correlated with the future exposure of a specific transaction. In such cases, the exposure tends to be highest precisely when the counterparty becomes riskier, creating a self-reinforcing risk profile.

From an operational standpoint, SWWR refers to situations where there is a direct and legal connection between the counterparty and the underlying asset – that is, cases where both belong to the same economic group and the relationship between default and exposure is effectively 100% correlated. These are known as Legally Connected (SWWR.LC) transactions.

Under the Internal Model Method (IMM), the treatment of SWWR is subject to specific regulatory constraints. The key requirements are:

- Exclusion from EPE calculation: exposures identified as subject to Specific Wrong-Way Risk must be excluded from the standard Expected Positive Exposure (EPE) calculation.
- Identification and documentation: institutions are expected to have internal procedures in place to clearly identify and tag transactions exposed to SWWR. In BBVA's current framework, SWWR.LC positions are flagged by the Global Market Risk Unit (GMRU) as part of this identification process.
- Capital treatment: once identified, SWWR.LC exposures are excluded from IMM and routed to the capital engine for capital requirement purposes under the Jump-to-Default (JTD) framework.

12. RNIEPEs

The Risks-Not-In-Expected-Positive-Exposure (RNIEPEs) framework is used to ensure that material risk drivers not explicitly captured in the expected positive exposure (EPE) measure—whether due to model simplifications, omitted risk factors, or structural limitations—are appropriately identified, assessed, and reflected in capital requirements.

Each identified RNIEPE must undergo a materiality assessment to determine whether it needs to be capitalised. For material RNIEPEs, a direct capital add-on should be applied, unless the risk is already incorporated conservatively in EEPE. For each RNIEPE, the corresponding Risk-Weighted Exposure Amount (RWA) can be computed using the following formula:

$$RWA_{RNIEPE} = \sum_n RW_n * \alpha * ERE_n$$

where RW_n denotes the risk weight corresponding to the n-th netting set affected by the RNIEPE and ERE_n is the corresponding expected-RNIEPE-exposure floored at zero.

In BBVA-IMM, the risk factor volatilities have been identified as a RNIEPE as they are not dynamically simulated in the effective EPE. To quantify this RNIEPE, an approach based on sensitivities is used where the volatilities are simulated up to 1Y as a multivariate gaussian with parameters based on historical data and correlated with the risk factors included in the model.

Spike-related RNIEPEs are required to be considered separately from other types of RNIEPEs due to their distinct nature and their potential to cause short-term but significant increases in exposure. An expected-spikes-exposure (ESE) only considering spikes is calculated as explained in [8.5 Spikes and Smooth exposure](#).

More details about the ERE calculation and the materiality assessment can be found in document:

[BBVA - RNIEPE Materiality Assessment](#)

13. CORE

CORE is the system used to calculate the official regulatory capital. The SACCR model is currently implemented here. In order to implement the IMM, due to Finance Department requirements to accommodate its reports, the CORE engine needs to receive some transaction-level metrics, and this requires performing a series of fictitious distributions. This document describes the entire implemented distribution logic.

[Cálculo del vencimiento efectivo en KCCR regulatorio, y reparto por operación para propósitos de reporting](#)

14. Expert criteria outside the model

This document's main purpose has been to specify the modelling assumptions and criteria of the IMM exposure model. However, there are some expert criteria, affecting the final exposure result, which are assumed to be defined outside of the scope of the risk model, due to them being more related to business processes or based on expert knowledge.

To ensure that these criteria are governed according to their specific nature, a different governance applies to them than what is the case for the IMM model, as defined in the Risk Models Inventory (although the regulatory IMM calculation could be said to encompass both). Changes will be approved according to a different committee structure, although, since they will affect the IMM, their

governance is also defined in the Rule for Changes of Rating Systems and IMA Models. Additional governance regarding these criteria is specified in the Risk Models Development Rule.

An inventory of current and past values of these criteria is maintained, and can be found here:

[Inventario Criterios Expertos No de Modelo](#)